REV 3 · M20 · AWAITING APPROVAL · 2026-08-07

`pretrain/final-dataset` — what we are building and why

For the whole team. Revision 3, 2026-08-07. Status: **plan awaiting owner approval. Nothing ingested.**
Branch `final-dataset` @ `origin/edullm/perf-threaded-verify-and-gatea` (0.9.1, the deployed code).

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1. The one-paragraph version

We are building a **1.0-trillion-token pretraining corpus** for **M20** — a **20.0B-total / 1.279B-active** Mixture-of-Experts model, education-focused, steered on MMLU, GSM8K, ARC and HellaSwag. The corpus ships in **two stages**, because no comparable published run used a flat mixture: a **900B bulk phase at 77% web**, then a **100B cooldown at 32% web** that concentrates math, code, QA and reasoning traces. Every token is headerless raw `uint32` little-endian under the **dolma2 tokenizer**, which we are keeping — that also makes AI2's 5.93T pre-tokenized corpus a byte copy rather than a re-tokenization. **No source exceeds 0.90 epochs**, so nothing repeats.

Why revision 3 exists. Revision 2 was written for a 40B/4B model at 1.3T with a *flat* 54% web share. All three of those numbers were wrong. §8 lists every claim that changed and why. Read it before relying on any figure here.

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2. The model, from the code rather than from a conversation

total params	**20,002,742,272**
active params	**1,279,369,216**
sparsity	**15.63 : 1**
shape	256 experts, **top-8**, **zero shared**, d_model 2048, L=24, GQA 16/4

Pinned at `OLMo-core/src/olmo_core/nn/transformer/config.py:1125` (`MAPLE_RUNGS["M20"]`, commented "*THE MISSION DELIVERABLE*") and `:1166`. `maple/HANDOFF.md`: "*Active is 1.279B, not 1.49B… **Plan against 1.279B.***"

This differs from how the model has been described verbally — "20B total / 2B active at 10:1" is **wrong by 56% on active params**. On the non-embedding convention most model cards use it is 0.868B active = 22.56:1. The 20.0B total is *correct* rather than an error: it differs from DeepGrove's published 20.2B/1.49B only by vocabulary, and the gap is exactly `2·2048·(151,936−100,352)`.

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3. Stage 1 — bulk, 900B (90% of budget), 77% web

source	share	tokens	pool	epochs	grade
DCLM-baseline	42%	378.0B	744.6B	0.51	MEASURED
FineWeb-Edu	28%	252.0B	1,583.1B	0.16	MEASURED
`common-pile/stackv2`	10%	90.0B	707.0B	0.13	DERIVED
FinePDFs-Edu	7%	63.0B	70.0B	0.90	CORRECTED
Nemotron-CC-Math-3+	5%	45.0B	133.0B	0.34	CARD ⚠️
FinePhrase (1 partition)	4%	36.0B	123.3B	0.29	MEASURED
academic (peS2o + PubMed + arXiv)	2%	18.0B	46.6B	0.39	MEASURED
reference (Wikipedia + pre-1929 books)	1%	9.0B	26.2B	0.34	MEASURED
StackExchange	1%	9.0B	25.9B	0.35	MEASURED

4. Stage 2 — cooldown, 100B (10% of budget), 32% web

source	share	tokens	pool	epochs
DCLM-baseline	32%	32.0B	744.6B	0.04
`common-pile/stackv2`	18%	18.0B	707.0B	0.03
Nemotron-CC-Math-3+	16%	16.0B	133.0B	0.12
AI2 dolma3 midtraining mix	14%	14.0B	100.0B	0.14
reasoning traces / worked examples	8%	8.0B	~50B	0.16
Cosmopedia	4%	4.0B	21.7B	0.18
Nemotron Math-Textbooks	3%	3.0B	27.5B	0.11
academic	3%	3.0B	46.6B	0.06
reference	2%	2.0B	26.2B	0.08

Combined: 1,000B, max epoch 0.90, nothing repeated. DCLM 41.0% · FineWeb-Edu 25.2% · code 10.8% · FinePDFs 6.3% · math 6.1% · synthetic 4.3% · everything else 6.3%.

FineWeb-Edu lineage is 28.8% — that matters because FinePhrase *is* rephrased FineWeb-Edu, so "edu-web" and "synthetic" are partly the same upstream pool. Revision 2 had this at **54.5%**: a quarter of the corpus was one pool entered twice under two labels. No published corpus does that.

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5. Why two stages, and why 77% then 32%

No comparable published run used a flat share. OLMo 2 §2.3 (arXiv 2501.00656, read directly): stage 1 is "*≥90% of training FLOPs*" and "*mostly web-sourced*"; stage 2 "*up-sample[s] the highest-quality web documents and curated non-web sources*" plus "*synthetic data crafted to patch math capabilities.*" OLMo 2 stage 1 is **95.1% web**, Olmo 3 **76.1%**, Olmix fixes web at **75%**; stage 2 drops to 27.5–52%.

Revision 2's flat 54% was therefore wrong in *shape*, not just level — about 25pp too low for the bulk and 20pp too high for the anneal.

And at our size the cooldown is where the metrics actually move. At 1B params, 4T tokens of 95%-web pretraining leaves MMLU near chance; mid-training moves MMLU **+17.4** and GSM8K **3.3 → 43.8**. That benefit is *largest* at our scale (**+37.0% at 1B vs +12.3% at 32B**). At 7B, across seven mid-training mixes, **MMLU spans 1.3 points while GSM8K spans 27.5**.

Web is DCLM-led rather than FineWeb-Edu-led, on direction only. DCLM's Table 8 is token-matched at 276B/7B and favours DCLM on MMLU — but hyperparameters were tuned on DCLM-baseline, and under FineWeb-Edu's own eval protocol the same paper says the two "*perform quite similarly.*" Nemotron-CC and SmolLM2 place FineWeb-Edu on **opposite sides** of DCLM. So we keep both, weighted toward DCLM, and treat the choice as unsettled. "**Which web" is worth ~16 MMLU at constant share — 4–8× any share effect** — which is why the split matters more than the percentage.

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6. QA and worked examples — first-class, and they pull in opposite directions

These were requested as explicit categories. The measured evidence says they must be **two** categories, not one. OLMo 3's compute-matched microanneals, each against a web-only control:

intervention	MMLU	ARC	GSM8K	Minerva	HumanEval
QA / MC-format (Table 8, 10B)	**+3.2**	**+2.6**	−1.2	−1.6	−1.5
reasoning traces (Table 9, 5B)	**−1.5**	+0.1	**+8.4**	**+7.3**	**+11.6**

They partially cancel on our four headline metrics. Any single "QA and worked examples" share would be arithmetically wrong. QA saturates at **~14% of the anneal** — AI2 gave it 13.9%, and beyond that the curve is flat (−0.1 / +0.7). Hence 14% dolma3-midtraining plus 8% reasoning traces in stage 2, scheduled separately.

Two honesty notes the team should carry.

The QA gain is format-matching, not knowledge. AI2's QA is GPT-4o-mini-rewritten *multiple-choice* from academic subreddits — format-matched to MMLU and ARC by construction. It does **not** transfer to StackExchange, which is prose Q&A. Our own P1 paper measured the ceiling directly: 65.24% under QA-shaped prompts collapses to **3.02%** chat-style and **0.00%** short-answer. P1 §2.6 lists format-independent knowledge as *not supported*.

The honest MMLU expectation from QA data is +0. The one experiment above the measurement threshold — Nemotron-CC at **8B/1T**, where MMLU is live at 48–53 — gives synthetic QA **+0.2 / −1.1**. The often-quoted Nemotron-CC "+5.6 MMLU" is *classifier selection*, not QA synthesis. And note the LR anneal alone gave **+2.0 MMLU** in OLMo 2, which is routinely misattributed to the data.

Worked examples buy math capability and cost something else, in two independent experiments. OLMo 3 at 7B: **+8.4 GSM8K, −1.5 MMLU**. Our own P1 at 370M: Pass@8 13.49 → 18.21, but **PassRatio@8 fell on every scaffolded arm** (5.11 → 4.33/4.52/4.59), and the pedagogically-ordered fade *lost* to random order. Both find capability up, reliability or knowledge down.

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7. Measurement — the constraint that would have voided our own experiments

Multiple-choice scoring is a separately-acquired skill and is random below ~400B tokens. OLMES (arXiv 2406.08446) Figure 1 caption, verbatim: "*During early training, there is good signal from CF while MCF is random.* ***Around 400B tokens, the model starts gaining the ability on the MCF format.***"

That sorts the literature into uninformative nulls and real zeros — and it invalidates part of our own plan, because **P1's probes ran 300M–10B tokens, far inside the dead zone.** Any MMLU column from a probe at that budget measures nothing.

So: score CF (cloze), or use bits-per-byte. P1 in fact used BPB — its 1.6062 vs 1.6329 figures are bits-per-byte, which is why that result stands. AI2 separately abandoned accuracy metrics at 1B/100B as "*too difficult to show improvement.*" The relationship **inverts** at the top end (93.7% MCF vs 69.0% CF on ARC-Challenge), so this is scale-dependent, not a blanket rule.

Gate on MMLU and GSM8K directly, at fixed intervals, and treat a knowledge regression as a stop condition. Knowledge collapse is the most reproducible harm in this literature — phi-4's 13.8×-synthetic arm cost **TriviaQA −14.8**; MoE data reuse took **MMLU 32.92 → 24.57**; Qwen2's 12T relaxed-quality run showed *no* improvement over 7T. **Repetition costs knowledge and spares reasoning**, which is exactly our weighted axis, and no loss- or BPB-based gate sees it.

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8. Contamination — read this before trusting any eval number

Seventeen corpora audited from papers and source code rather than dataset cards. **Nobody earns a TRUST verdict.** Two findings are directly actionable:

1. **Nemotron-CC-Math's decontamination gate effectively never fired.** An independent 13-gram scan found **11,868 contaminated documents remaining against MATH500 alone — 13.2× more than NVIDIA's entire removal budget** — including verbatim GSM8K test items at Jaccard 1.0. This does not change the pillar choice (every alternative scores worse) but it makes mitigation mandatory.
2. **The Common Pile ships GSM8K in Flan CoT format** (`fc-cot-cot_gsm8k`, 6 repeats, ~9× cooldown upweight). AI2 dropped Winogrande *the benchmark* rather than the data, and GSM8K was not in their suite so it never triggered the reflex. **Fix: drop one source, 0.51% of tokens.**

tier	action	cost
1	**exclude 9 items at source — carries nearly all the value**	**<1% of tokens**
2	chunked embedding cos≥0.5 → LLM judge, on the anneal + synthetic + question-bank slices	\$500–2,000
3	13-gram as an **instrument**, not a filter — publish the number	\$10–50
4	**do not do** — including our own previously-decided `decon --ngram_size 5`	—

Three things everyone should know: **a quality classifier is a contamination amplifier** (the DCLM classifier puts *all* MMLU and GSM8K needles above the 99th percentile — and DCLM is now our largest source); **13-gram matching scores F1 0.926 on verbatim text but exactly 0 on rephrased text**, so our synthetic slice is structurally un-decontaminatable by n-grams; and **HellaSwag is negative in four independent papers** while being where contamination persists longest — our least-served and least-instrumented metric from both directions.

9. Build and cost

Tokenizer: dolma2, kept. No alternative clears zero measured downstream gain — SuperBPE is a null at MoE scale with a code regression, a bigger vocab is contradicted by the resolved scaling law, and `dolma2_sigdig` is a 29-minute abandoned experiment AI2 never trained on. Keeping dolma2 also means **AI2's pre-tokenized shards are byte-identical to our format** (verified by range-read: no NumPy magic, valid ids from byte 0), so their corpus is a copy.

Shards: 50,003,968 tokens → ~20,000 objects. Both earlier objections were wrong: the 8192 alignment is *not* mandatory (the check is skipped unless a group declares `seq_len`, and ours does not), and mixture error is bounded by `1/shards_per_component` — **0.33% worst case, not 4.8%.**

⚠️ **H100 is not provisioned.** Live compute profiles are T4 / L4 / L40S / A10G / `gpu-8xa100`; both H100 shapes went unprovisioned after 6,815 capacity failures. **Every 400 TFLOP/s figure in our research reports is 3.2–4.3× optimistic.**

	8×A100 (today)	8×H100 (if unblocked)
1.0T tokens	**$69.9k / 88.8 days**	**$36.7k / 27.8 days**

Unblocking H100 is worth more than any budget decision available to us.

The one genuinely MoE-specific requirement: micro-batches must not be domain-pure. Load-balancing loss computed over a domain-pure micro-batch forces uniform routing and suppresses expert specialization — worth **0.13–0.18 PPL and +5–6 GSM8k**, more than a 50% increase in activated FLOPs buys. Our shards are per-source, so every micro-batch is domain-pure by construction, and `MoELoadBalancingLossGranularity` defaults to `local_batch` in four places. At M20's 15.63:1 with zero shared experts this is documented as routing **collapse**, not merely suppressed specialization. It must be fixed **before** training: switching at 10% of the run recovers only ~55%.

Everything else about the corpus we choose exactly as for a dense model — Qwen3 published that a MoE reaches dense performance *on the same data*.

The one unbackfillable decision: store a per-document quality percentile as a label, at ingest. Every 2026 corpus reweights on per-document quality and topic labels. `entry.labels` sits inside `manifest_sha256`, so this cannot be added later without re-copying the corpus. A wrong budget is a config field; a wrong schema is a v2.

10. What changed since revision 2, so you can calibrate

claim	correction
"40B total / 4B active"	**20.0B / 1.279B**, from the code
"1.3T budget"	**1.0T**, on Ling-mini-beta parity (active params match to 1.6%, d_model identical, measured parity with a dense 6.11B at exactly 1T)
"flat 54% web"	**two stages, 77% then 32%**
"8192 alignment is mandatory"	Conditional on a `seq_len` our corpus never declares
"50M shards cost 2.4→4.8% error"	Off by ~15×. Real bound 0.33%. **This was used to justify a decision**
"code caps at 74.81B"	`common-pile/stackv2` was never checked. **~707B**
"FineWeb authors recommend 50% FinePDFs"	Traces to a card saying "*inspired by*" an unaffiliated blog using a **70M-param GPT-2**. Their real guidance is "*keep PDF below 25%*"
"NVIDIA replicates DCLM > FineWeb-Edu"	Those deltas appear **nowhere** in that paper; its controlled test has **FineWeb-Edu winning**
"MegaMath: 18× the data, worse result"	The anneal **is** token-matched; my framing was backwards
"min_ngram 13→5 costs −11.8 MMLU"	A **deduplication** result, not decontamination
"77.5% Common Crawl is a problem"	Normal — Llama 1 was 82%, dolma3 76%
"Aryabumi brackets our scale"	Its code sweep ran **only at 470M**

Still unverified, and flagged rather than used: Gate A's cost at ~20,000 objects (our own extrapolation, never tested); Nemotron-CC-Math's 133B (card-grade); FinePDFs-Edu's ~70B (byte-derived, after the recorded 161.07B proved wrong by 2.3×); MegaMath-Web's post-filter pool size; and Dolma 3's adult-content prevalence, which is **blocking** for that source.

The largest residual risk: every mixture study cited in our seventeen research reports is DENSE. No mixture ablation on a sparse MoE exists at any scale. Transfer to M20 is unverified, and that unknown is larger than any error listed above.

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11. What happens next

1. **Wire `keeps_id` into the build path.** It exists, is tested on 287,000 ids, and is called from nothing that writes data — so today's declared synthetic rests on ~28% as many real documents.
2. **Fix `MoELoadBalancingLossGranularity`** (§9) and the **missing EOS** in `gigatoken-superbpe`, where `added_tokens: []` makes Gate A's EOS check skip silently in two published corpora.
3. **Drop `data_provenance_initiative`** — 0.51% of tokens, removes GSM8K-in-Flan-CoT.
4. **Decide the label schema** (§9, unbackfillable).
5. **Measure the three blockers:** Dolma 3 adult content, Nemotron-CC-Math contamination post-fix, and a real dolma2 count for Nemotron-CC-Math.
6. **Then ingest**, source by source, each as a separately-approved platform job.

Every AWS job goes through the platform form and needs a human release. Nothing auto-publishes.